MIL-M-38795B(USAF)
19 August 1982
Superseding
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9 April 1973

MILITARY SPECIFICATION

MANUAL, TECHNICAL: SYSTEM PECULIAR CORROSION CONTROL PREPARATION OF

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for preparation of technical manuals which provide instructions and guidance to organizational, intermediate, and depot level maintenance personnel for system peculiar corrosion prevention and control on aircraft and missiles. With some modifications, it may serve as a guide for system peculiar corrosion control manuals for other types of equipment such as ground support equipment and vehicles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

MILITARY SPECIFICATIONS

MIL-M-38784	Manuals, Technical: General Style and Format Requirements
MIL-P-38790	Printing Production of Technical Manuals: General Requirements for

MILITARY STANDARDS

MIL-STD-1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapons Systems
MIL-STD-1587	Materials and Processes Requirements for Air Force Systems

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document shall be addressed to: HQ AFLC/LOLME, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Technical Orders

1-1-1	Cleaning of Aerospace Equipment
1-1-2	Corrosion Prevention and Control for Aerospace Equipment
1-1-3	Preparation, Inspection and Repair of Aircraft Fuel, Oil and Water Alcohol Cells and Integral Tanks
1-1-4	Exterior Finishes, Insignia and Markings Applicable to USAF Aircraft.
1-1-8	Application of Organic Coatings, Aerospace Systems
1-1-689	Prevention and Control of Corrosion and Fungus in Communication, Electronic, Meteorological and Avionic Equipment

(Copies of documents required by Contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preparation: The general manner of preparation of the manuals shall be in accordance with MIL-M 38784 and MIL-P-38790.

3.2 Contents: The manual shall contain that information required by maintenance personnel for determining the location and extent of corrosion damage to a particular system, and instructions for its removal and treatment. The manual shall not contain information already included in general series technical orders pertaining to corrosion control except to the extent essential for continuity of the presentation of system peculiar technical data. Specific instructions in general series technical orders shall be referenced where required throughout the manual.

3.2.1 Corrosion removal and prevention procedures. When corrosion removal and prevention procedures covered in the general series technical orders are sufficient for the applicable aerospace equipment, it shall be so stated by reference in the appropriate section(s) of the system peculiar manual. If the information in the general series technical orders is not sufficient, the system peculiar manual shall contain the necessary information to augment the general series technical orders.

3.2.2 New materials. When new materials of construction are used in the original fabrication of the system/end item, an alternate (substitute) material, shall be identified for replacements, if available and applicable. The alternate material shall be drawn from those standard materials listed in the general series technical orders, MIL-STD-1568, MIL-STD-1587, or other approved government specifications.

3.2.3 Special or proprietary equipment. Use of "special" or proprietary equipment shall not be specified in the system peculiar technical manuals without prior approval of the procuring activity and the Air Force Corrosion Program Manager, WR-ALC/MMEMC.

3.2.4 System peculiar manual. The system peculiar manual shall not refer to any contractor's process or materials specifications since they are not available for use by field activities. The applicable detailed process instructions shall be specified in the system peculiar manual if not covered by Government specifications for reference.

3.2.5 Precautions. Precautions concerning the use of toxic materials and hazards associated with their use shall be included. Special instructions.

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warnings, etc., shall be specified. Methods of disposal shall be specified when required. Precautions for working in hazardous areas (propellant areas, fuel areas, etc.) shall be noted when required.

3.3 Arrangement. The manual shall include in the following order: front matter; a list of relevant publications; an introduction; Section 1, Cleaning and temporary protection; Section 2, Paint removal and painting; sections on sealing; special sections if needed; sections on corrosion prone areas; and a special equipment section. A glossary and index will be included if required by MIL-M-38784. An appendix may be included if needed.

3.3.1 Front matter. The title page, list of effective pages, table of contents, and list of illustrations shall be prepared in accordance with the requirements of MIL-M-38784.

3.3.2 List of relevant publications. This shall include the designation and title of general series TOs: TO 1-1-1, TO 1-1-2, TO 1-1-3, TO 1-1-8, and TO 1-1-689. For aircraft, it shall include TO 1-1-4. All other relevant manuals (general series and system peculiar TOs) shall be listed.

3.3.3 Introduction. The introduction shall contain an explanation of the purpose of the manual and shall include any relevant information that will be of assistance in the use of the manual. It shall briefly describe the information contained in the manual.

3.3.4 Sections

3.3.4.1 Section I - Cleaning and temporary protection. This section shall cover cleaning and/or washing procedures for the system and procedures for use of corrosion preventive compounds for temporary protection of specific areas after washing where needed.

3.3.4.2 Section II - Paint removal and painting section. This section shall give precautions and procedures, which are applicable to the specific system and are not adequately covered in TO 1-1-8. Example: masking diagrams, etc. General corrosion removal and treatment procedures may be included in this, or another section, if the procedures are not sufficiently covered in TO 1-1-2 or TO 1-1-689.

3.3.4.3 Section on sealing. A section describing types of sealants used and sealing procedures shall be included and contain references to TO 1-1-2 or TO 1-1-3 where applicable.

3.3.4.4 Special sections. Special sections (such as a section on the control of microbial infestation in fuel tanks) may be included if a special problem exists, or if the special procedures for inspection and treatment are not covered in the general series TOs.

3.3.4.5 Sections on corrosion prone areas. For aircraft, and larger/medium size missiles, and equipment, there will be a separate section for each major structural group, auxiliary system or major area. Example: wings, fuselage, empennage, landing gears and system components. These sections will discuss each corrosion prone area separately and present the following information for each area:

a. Photographs (color, if required) and line type drawings shall be used to illustrate areas and conditions and shall be placed on a page facing the text for each corrosion prone area.

b. A paragraph or paragraphs titled "General" with the following information: A discussion of the types of corrosion which may be found in the area, and where it is likely to occur. The causes of the commonly found corrosion in each area will be discussed. The metal and alloy types will be discussed. Any operational changes in the system which may significantly affect corrosion

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will be discussed. This shall include changes in alloys, sealing materials, finishes, etc.; and any other general information about this area. Serial numbers of aircraft affected by production changes shall be included.

c. A paragraph or paragraphs shall discuss the classification of structure in the area as to what is critical, non-critical, etc. Any necessary warning for critical structure shall be given.

d. A paragraph giving damage limits or a reference to where the damage limits are located.

e. A paragraph giving the preferred inspection method to include reference to the Non-Destructive Inspection (NDI) manual.

f. A paragraph or paragraphs giving the recommended treatment/corrosion removal procedures.

g. A paragraph or paragraphs to identify the type of sealant and sealing methods to be used with suitable reference to other parts of the manual, TO 1-1-2, 1-1-3, and 1-1-689 as required.

h. A paragraph or paragraphs to identify the finish system to be used and the procedures for application with suitable reference to TO 1-1-3, 1-1-8, and 1-1-689 as required

3.3.4.6 Special equipment and materials section. This section shall contain a short introduction, a table of any special equipment cited in other sections of the manual, and a table of any special materials cited in other sections of the manual and shall be the last section in the manual. Other manuals for the system, such as the -2 TO may be referenced in lieu of listing materials and equipment in these tables. Materials and equipment listed in TO 1-1-2 tables shall not be listed in the manual tables.

3.3.5 Illustrations and tables.

3.3.5.1 Illustrations and tables in sections on corrosion prone areas. In addition to the requirements of 3.3.4.5, the following illustrations and tables shall be included at the end of every section of the manual on corrosion prone areas (i.e. wing, fuselage, etc.).

3.3.5.1.1 Illustrations. An illustration or group of illustrations shall be provided which identify each of these areas/items separately; (1) magnesium components, (2) critical forgings and/or expensive and difficult to replace components (i.e., primary structure), (3) high strength steel (over 180 KSI) components, and (4) areas subject to collect moisture and area drain locations (condensate and urinal drains, and battery vents). These items/areas must be inspected and protected. An arrow should be included on structural illustrations to show the grain flow. The area of a component which has a tendency to corrode should be highlighted and a location key used to coincide with a key in the text of a corrosion prone area and in the table of corrosion tables specified below

3.3.5.1.2 Corrosion inspections. A table on corrosion inspections shall follow the format in figure 1, with the following information in a separate column:

a. Index column - This column shall contain an index number which will be the key in the illustration and text. Critical areas shall be identified with a "C" before the index column.

b. Area/Item column - This column shall give the [illegible] and nomenclature of the corrosion prone area.

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c. Material identification - This column shall list the basic metal followed by the standard alloy code in parentheses. When the material has a special surface treatment such as shot peen, stress roll, etc., it shall be specified. Any material change will be noted.

d. Corrosion appearance - This column shall identify the type of corrosion anticipated such as oxidation, pitting, galvanic, etc. A detailed description of what the corrosion will look like on each specific part, based on the use location/exposure of the part shall be included in this column or a reference to the discussion of known corrosion problems in the section shall be given. General statements such as "red rust" for steel parts or "white or grey powder" for aluminum parts are not adequate. Specific locations on the part where corrosion may be expected to start shall be identified.

e. Preferred inspection method - This column shall give the preferred inspection method in brief form. Long procedures shall be included in the manual discussion and shall be referenced by the table. NDI procedures shall not be given in detail in the discussion unless they cannot be referenced in another Air Force manual (TO).

f. Probable cause - This column shall give the probable cause of the corrosion normally encountered in the area.

3.3.5.1.3 Corrosion removal and repair procedures. A table giving the corrosion removal and repair procedures will be keyed to the illustrations and text for the corrosion prone areas. It will follow the format in figure 2. The table will consist of the following columns:

a. Index - Same as figure 1. See 3.3.5.1.2.a

b. Area/item and nomenclature - Same as figure 1. See 3.3.5.1.2.b.

c. Damage limit - This column shall provide the damage limits. It shall give specific grind-out limits for each area/item.

d. Repair - This column shall specify the corrosion repair procedure or reference a procedure in a general series TO, in the text for the corrosion prone area in the section, or in another system peculiar technical order. Any material change will be noted with the procedure for the new material being listed also.

3.3.5.1.4 Sealant repair table. A sealant repair table shall follow the format in figure 3. This table shall give the sealant repair procedure for all areas of the equipment covered by a section with area/reference key for a specific corrosion prone area being used where applicable. Illustrations shall be used if needed. Detailed procedures in TO 1-1-2, TO 1-1-3 or the sealing section of the manual shall be referenced.

3.3.5.1.5 Finish system table and illustrations. Illustrations and a table in each section shall show every type coating required for all areas of the equipment covered by the section with exact areas being defined. The table shall follow the format in figure 4 and shall be keyed to the corresponding illustration or illustrations. The area/reference key for a specific corrosion prone area will be used in the table also where applicable. Both interior and exterior areas shall be shown. Finish system application procedures in TO 1-1-2, TO 1-1-8, and/or the paint removal and painting section of the manual will be referenced in the "touch up repair or repaint" column.

3.3.5.2 Table of special equipment. This table shall follow the format in figure 5 and shall list any special equipment necessary for corrosion inspections and control in accordance with the manual.

3.3.5.3 Table of special materials. This table shall follow the format in figure 6 and shall list any special materials necessary for corrosion control in accordance with the manual.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. The quality assurance provisions of MIL-M-38784 and MIL-P-38790 are applicable.

4.2 Coordination. Prior to publication, the proposed manual will be coordinated with WR-ALC/MMEMC and the AFLC System Manager.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing, and marking for shipment shall be in accordance with MIL-M-38784 and MIL-P-38790.

6. NOTES

6.1 Intended use. The technical manuals covered by this specification are intended to supplement the general series technical orders in solving system peculiar corrosion problems.

6.2 Definitions.

6.2.1 General series technical orders. Technical orders which contain general information applicable to all type systems. Example" TO 1-1-1, TO 1-1-2, TO 1-1-3, etc.

6.2.2 System peculiar. The term "system peculiar" indicates that information is applicable to one and only one weapon system or other item of aerospace equipment, but all series of a system may be included.

6.2.3 Special materials. Special materials are those corrosion control materials not listed in the general series technical orders that are required for use on a particular aerospace system.

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.	Option
3.2	Technical Orders	DI-M-3407	-----

(Data item descriptions related to this specifications, and identified in Section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data items descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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method

* Critical area

FIGURE 1. Corrosion inspection format.

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TABLE II. Corrosion removal and repair.

Index No. *	Area/Item nomenclature	Damage limit	Repair procedure

* Identical area

FIGURE 2. Corrosion removal format.

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TABLE III. Seala... system.

Index No. *	Area/Item nomenclature	Repair procedure

* C-Critical area

FIGURE 3. Sealant system format.

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TABLE IV. Finish system.

Index no.	Area/Item and nomenclature	Metal identification	Original pretreatment	Color		Touch-Up repair or repaint		
				Name	Number (F.S. 595)	Pretreatment	Primer	Top coat

FIGURE 4. Finish system format.

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TABLE V. Special equipment list.

Item	Specification (If Available)	Stock number or part no.	Source	Intended use
			(List only when there is only one)	

FIGURE 5. Special equipment list format.

TABLE VI. Special materials. (consumable)

Item	Specification	National stock number	Unit of issue	Source	Intended use
				(List only when there is only one)	

FIGURE 6. Special materials format (consumable).

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